Homework 7

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MATH 421 — The Theory of Single Variable Calculus
Section 003
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Collaboration Statement

I went to course assistant 5:30-7pm on 10/30 namely to get a direction/high level overview on how to do problem 6 because I was high key overthinking it. All written work and final write-up are my own.

Problem 1 *(Ross Chapter 2 11.4)*

Repeat Exercise 11.2 for the sequences:
$w_n = (-2)^n$, $x_n = 5^{(-1)^n}$, $y_n = 1 + (-1)^n$, $z_n = n\cos(\frac{n\pi}{4})$.

Question 11.2 asks you to do the following for each sequence:

(a) For each sequence, give an example of a monotone subsequence.

(b) For each sequence, give its set of subsequential limits.

(c) For each sequence, give its $\limsup$ and $\liminf$.

(d) Which of the sequences converges? diverges to $+\infty$? diverges to $-\infty$?

(e) Which of the sequences is bounded?

Discussion.
So we will analyze each sequence one by one, finding monotone subsequences (which we know must exist by 11.4 Theorem), subsequential limits, lim sup and lim inf, convergence/divergence, and boundedness for each.

So from textbook, we know that a *monotone subsequence* is a subsequence that is either nonincreasing or nondecreasing.

A *subsequential limit* is a limit of some subsequence of the original sequence.

The lim sup is the supremum (least upper bound) of the set of subsequential limits, and the lim inf is the infimum (greatest lower bound) of the set of subsequential limits.

We also have a ton of helpful theorems to use from the textbook, including:

- **11.2 Theorem**: Let (s_n) be a sequence.
  (i) If $t \in \mathbb{R}$, then $\exists$ a subsequence of (s_n) that converges to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for every $\epsilon > 0$.
  (ii) If (s_n) is unbounded above, it has a subsequence with limit $+\infty$.
  (iii) If (s_n) is unbounded below, it has a subsequence with limit $-\infty$.

  In either case, the subsequence can be taken to be monotonic.
- **11.3 Theorem**: If the sequence (s_n) converges, then every subsequence of (s_n) converges to the same limit.
- **11.4 Theorem**: Every sequence has a monotone subsequence.
- **11.5 Theorem (Bolzano-Weierstrass Theorem)**: Every bounded sequence has a convergent subsequence.
- **11.7 Theorem**: Let (s_n) be a sequence. There exists a monotonic subsequence whose limit is $\limsup s_n$, and there exists a monotonic subsequence whose limit is $\liminf s_n$.
- **11.8 Theorem**: Let (s_n) be a sequence in $\mathbb{R}$, let S be the set of subsequential limits of (s_n).
  (i) $S \neq \emptyset$
  (ii) $\sup S = \limsup s_n$ and $\inf S = \liminf s_n$
  (iii) $\lim s_n$ exists $\iff$ S contains exactly 1 element (namely $\lim s_n$)
- **11.9 Theorem**: Let S denote the set of subsequential limits of a sequence (s_n). Suppose (t_n) is a sequence in $S \cap \mathbb{R}$ and that $t = \lim t_n$. Then $t \in S$.

  TL;DR: The set of subsequential limits is closed.

  So with all this in the back of our minds (or in front of my face because I have not yet fully baked these theorem in my head), we can use this to help find subsequential limits for each sequence.

Answers.
For $w_n = (-2)^n$, we know the sequence follows the pattern: $(-2, 4, -8, 16, -32, 64, \dots)$ whose even terms diverge to $+\infty$ and odd terms to $-\infty$ as n increases. With this in mind:

(a) Monotone subsequence of w_n could be the even terms: $(4, 16, 64, \dots)$ which is increasing wherein the selector function could be $\sigma(k) = 2k$ for $k \in \mathbb{N}$. The other monotone subsequence could be the odd terms: $(-2, -8, -32, \dots)$ which is decreasing wherein the selector function could be $\sigma(k) = 2k - 1$ for $k \in \mathbb{N}$.

(b) The set of subsequential limits is $\{+\infty, -\infty\}$ since we have subsequences that diverge to both $+\infty$ and $-\infty$.

(c) $\limsup w_n = +\infty$ and $\liminf w_n = -\infty$.

(d) The sequence diverges to both $+\infty$ and $-\infty$.

(e) The sequence is unbounded.

For $x_n = 5^{(-1)^n}$, we know the sequence follows the pattern: $(5^{-1}, 5^1, 5^{-1}, 5^1, \dots)$ or $(\frac{1}{5}, 5, \frac{1}{5}, 5, \dots)$. With this in mind:

(a) Monotone subsequence of x_n could be the odd terms: $(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \dots)$ which is constant (and thus nondecreasing) wherein the selector function could be $\sigma(k) = 2k - 1$ for $k \in \mathbb{N}$. The other monotone subsequence could be the even terms: $(5, 5, 5, \dots)$ which is constant (and thus nondecreasing) wherein the selector function could be $\sigma(k) = 2k$ for $k \in \mathbb{N}$.

(b) The set of subsequential limits is $\{\frac{1}{5}, 5\}$ since we have subsequences that converge to both $\frac{1}{5}$ and 5.

(c) $\lim\ sup x_n = 5$ and $\lim\ inf x_n = \frac{1}{5}$.

(d) The sequence diverges.

(e) The sequence is bounded.

$\square$

Problem 2 *(Ross Chapter 2 11.8)*

Use Definition 10.6 and Exercise 5.4 to prove $\liminf s_n = -\limsup(-s_n)$ for every sequence (s_n).

Discussion.

10.6 Definition:

Let (s_n) be a sequence in $\mathbb{R}$. We define:

$$\limsup s_n = \lim_{N\to\infty} \sup\{s_n : n > N\}$$

and

$$\liminf s_n = \lim_{N\to\infty} \inf\{s_n : n > N\}$$

5.4 Exercise:

Let S be a nonempty subset $\mathbb{R}$, and let $-S = -s : s \in S$. Prove $\inf S = -\sup(-S)$. *Hint: For the case $-\infty < \inf S$, simply state that this was proved in Exercise 4.9.*

Proof.

From the definition of 10.6, we have:

$$\liminf s_n = \lim_{N\to\infty} \inf\{s_n : n > N\}$$

where we describe the set $S_N = \{s_n : n > N\}$ such that we can rewrite the expression for $\liminf(s_n)$ as:

Thus, we have:

$$\liminf s_n = -\limsup(-s_n)$$

which applies for every sequence (s_n) since we know the result from Exercise 5.4 holds for any nonempty subset of $\mathbb{R}$, and thus every sequence. $\square$

Problem 3 *(Ross Chapter 2 11.10)*

Let (s_n) be the sequence of numbers in Fig. 11.2 listed in the indicated order.

(a) Find the set S of subsequential limits of (s_n).

(b) Determine $\limsup s_n$ and $\liminf s_n$.

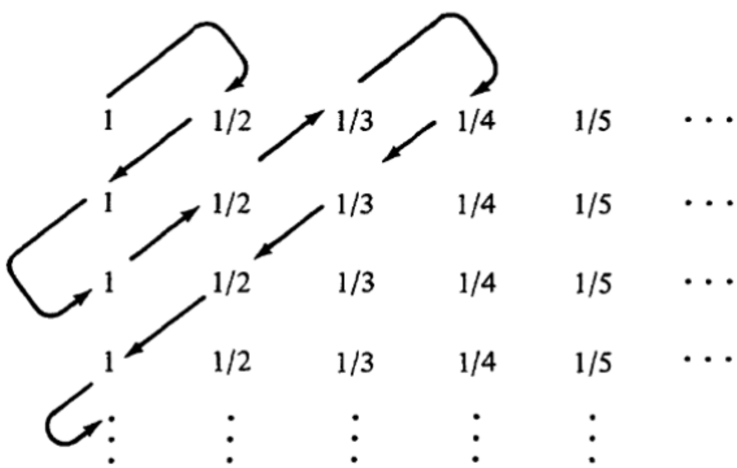

FIGURE 11.2

Discussion.

Proof.

$\square$

Problem 4 *(Ross Chapter 2 12.2)*

Prove $\limsup |s_n| = 0$ if and only if $\lim s_n = 0$. **Discussion.**

Proof.

$\square$

Problem 5 *(Ross Chapter 2 12.3)*

Let (s_n) and (t_n) be the following sequences that repeat in cycles of four:

$$(s_n) = (0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, ...)$$

$$(t_n) = (2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, ...)$$

Find

(a) $\liminf s_n + \liminf t_n$

(b) $\liminf(s_n + t_n)$

(c) $\liminf s_n + \limsup t_n$

(d) $\limsup(s_n + t_n)$

(e) $\limsup s_n + \limsup t_n$

(f) $\liminf(s_n t_n)$

(g) $\limsup(s_n t_n)$

Discussion.

Proof.

□

Problem 6 *(Ross Chapter 2 12.10)*

Prove (s_n) is bounded if and only if $\limsup |s_n| < +\infty$.

Discussion.

Proof.

□